INCH - POUND

MIL-PRF-32049 (AR)
23 September 1999

PERFORMANCE SPECIFICATION

PALLET, FIELD ARTILLERY PROJECTILE

This specification is approved for use by the US Army Armament, Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for three types of pallets used for storing and transporting 155MM projectiles. (see 6.1).

1.2 Classification. Classify the field artillery projectile pallet as follows:

TYPE I - Field Artillery Projectile Pallet with chemical agent resistance, increased durability, reusability and non-combustible which meets all requirements of this specification for a single length projectile.

Style A – M107 Family
Style B – M795 Family
Style C – M549 Family
Style D – M864 Family

TYPE II - Field Artillery Projectile Pallet which meets all requirements of this specification for all length projectiles, except 3.3.3, 3.3.4, 3.4.7, 3.5.2, and 3.5.4. Assemble this pallet with steel strapping.

TYPE III - Field Artillery Projectile Pallet with chemical agent resistance, increased durability, reusability and non-combustible which meets all requirements of this specification for all length projectiles.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armaments Research, Development and Engineering Center, ATTN: AMSTA-AR-QED, Picatinny Arsenal, NJ 07806-5000 or by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or letter.

AMSC N/A FSC 8140
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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

FEDERAL

FED-STD-595	Colors Used In Government Procurement
FED-STD-101	Test Procedures for Packaging Materials

MILITARY

MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-1916	DoD Preferred Methods for Acceptance of Product

SPECIFICATIONS

MILITARY

MIL-PRF-63460	Lubricant, Cleaner and Preservative for Weapons and Weapons Systems

(Unless otherwise indicated, copies of the above specifications, standards and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation. See section 6.5.

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DRAWING

US ARMY TACOM-ARDEC

8794342	Seal, Metallic, Assembly, Details and Instruction for Application

(Copies of this drawing are available from US TACOM -ARDEC, AMSTA-AR-QAD, Picatinny Arsenal, NJ 07806-5000)

SPECIFICATION

A-A-208	Commercial Item Description Ink, Marking, Stencil, Opaque (Porous and Non-Porous Surfaces)

(Copies of this specification are available from Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094)

US ARMY DEFENSE AMMUNITION CENTER AND SCHOOL (USADACS)

19-48-4076	Bundling of Separate Loading Projectiles for Storage and Transport

(Copies of the above drawing are available from US Army Defense Ammunition Center, 1 C Tree Road, Bldg. 35, McAlester, OK 74501-9053)

PUBLICATIONS

INTERNATIONAL TEST OPERATING PROCEDURES

ITOP 1-2-601	Laboratory Vibration Schedules dated 1 March 1988

(Copies of the above publication are available from US Army Combat System Test Activity, STECS-AD, Aberdeen Proving Ground, MD 21005-5059)

TEST OPERATING PROCEDURES

TOP 8-2-111	NBC Contamination Survivability, Small Items of Equipment

(Copies of the above publication are available from US Army Dugway Proving Ground, STECS-MT-TM-CB, DUGWAY Proving Ground, Dugway UT 84022-5000)

EVT-TP-1-86	Transportability Testing Procedures

(Copies of the above publication are available from US Army Defense Ammunition Center and School, SIOAC-DET, Savanna, IL 61074-9639)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B93.63M	Hydraulic Fluid, Petroleum
ANSI K62.344	Nithiazine Insecticide
ANSI/SAE AMS 3054A	Oil, Lubricating, Low temperature

(Address applications for copies to American National Standards Institute, ATTN: Customer Service, 11 West 42nd Street, New York, NY 10036 or http://www.ansi.org)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B117-95	Operating Salt Spray (Fog) Apparatus
ASTM E162-94	Surface Flammability of Materials Using a Radiant Heat Energy Source
ASTM D235	Standard Specification for Mineral Spirits (Petroleum Spirits) (Hydrocarbon Dry Cleaning Solvent)
ASTM D610	Evaluating Degree of Rusting on Painted Steel Surfaces
ASTM E662-95	Specific Optical Density of Smoke Generated by Solid Materials
ASTM E800-95	Measurement of Gases Present or Generated During Fires
ASTM D1655	Standard Specification for Aviation Turbine Fuel
ASTM D3951	Standard Practice for Commercial Packaging
ASTM D4175	Standard Terminology Relating to Petroleum, Petroleum Products and Lubricants

(Address applications for copies to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959 or http://www.astm.org)

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SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE J 300	Engine Oil Viscosity Classification
SAE J 312	Gasoline, Automotive, Combat
SAE J 313	Fuel, Diesel, Grade DF-2
SAE J 1034	Antifreeze, Ethylene Glycol
SAE J 1447	Hydraulic Fluid, Fire-resistant

(Address applications to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096-0001 or http://www.sae.org)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.

3.2 Design, materials, and manufacturing processes. Unless otherwise specified, the selection of the design, materials, and manufacturing processes is the prerogative of the contractor as long as all articles submitted to the government fully meet the operating, interface, ownership and support, and operating environment requirements specified.

3.3 Operating requirements. Each pallet shall meet the following functional, operational, and performance requirements.

3.3.1 Projectile storage. TYPE I and TYPE III pallets shall be capable of carrying and storing from two (2) to eight (8) 155MM artillery projectiles of the same length. TYPE II shall be capable of carrying and storing eight (8) rounds only. TYPE II pallets shall be capable of being strapped with 1 ½ inch steel banding. TYPE III pallets shall be capable of carrying and storing all lengths of projectiles in 3.3.1.1. When the pallet is completely closed, the projectiles shall be maintained on the pallet in a vertical orientation, show no noticeable lateral movement, nor allow the release of any of the projectiles, throughout the full operating environment spectrum. The pallet shall provide adequate clearance around the projectile to allow removal and attaching of grommets or flexible rotating band covers. The pallet shall hold the projectiles at the lifting plug and the boattail. The pallet shall have a means to prevent metal to metal abrasion/corrosion between the projectile boattail and the pallet base if a metal pallet is used. When the pallet is empty, the pallet shall be able to be disassembled to minimize space.

3.3.1.1 Projectile lengths and dimensions. The interface drawings listed in Table I give the length of the projectile, the dimensions of the projectile boattail, the dimensions of the grommet and the dimensions of the lifting plug. These interface dimensions are given to insure that the pallet properly interfaces with the projectile body, boattail and lifting plug.

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TYPE I: The pallet shall be capable of meeting all requirements when storing projectiles from one of the lengths given in Table I below.

TYPE II: The pallet shall be capable of meeting all requirements (except 3.3.3, 3.3.4, 3.4.7, 3.5.2 and 3.5.4) when storing all projectiles lengths given in Table I below.

TYPE III: The pallet shall be capable of storing all projectile lengths given in Table I below.

Table I. Interface drawings

Projectile Family	Projectile Body Approximate Length	Projectile Interface Drawings
Style A - M107	23.8 inches	Figures 1, 2, 3.
Style B - M795	29.5 inches	Figures 4, 5, 6.
Style C - M549	30.6 inches	Figures 7, 8, 9.
Style D - M864	31.6 inches	Figures 10, 11, 12.

The projectiles have a grommet or flexible rotating band cover (FRBC) attached. The projectile interface drawings showing length, boattail dimensions, lifting plug and grommet or FRBC are given in the table above.

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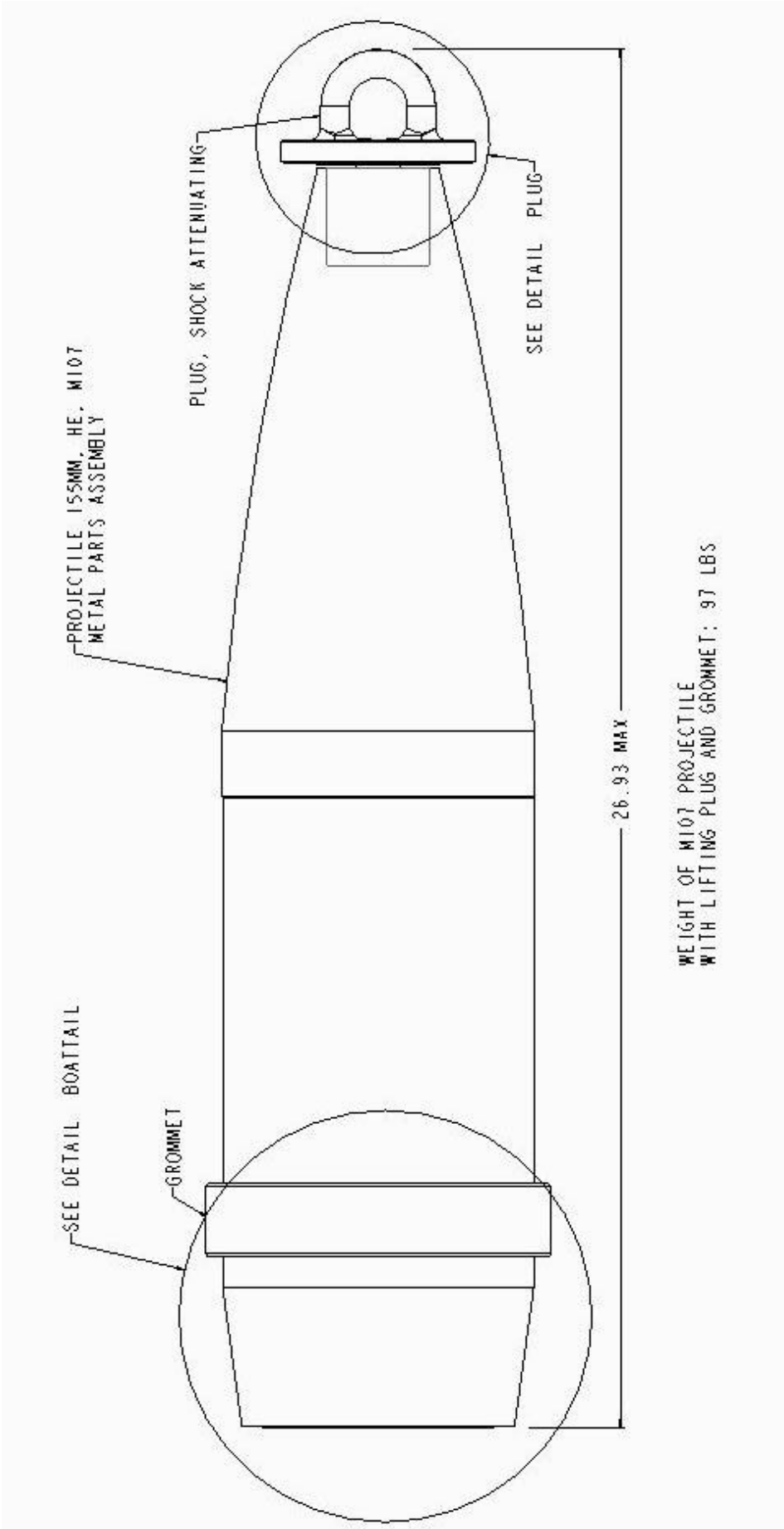


Figure 1. M107 Interface Drawing

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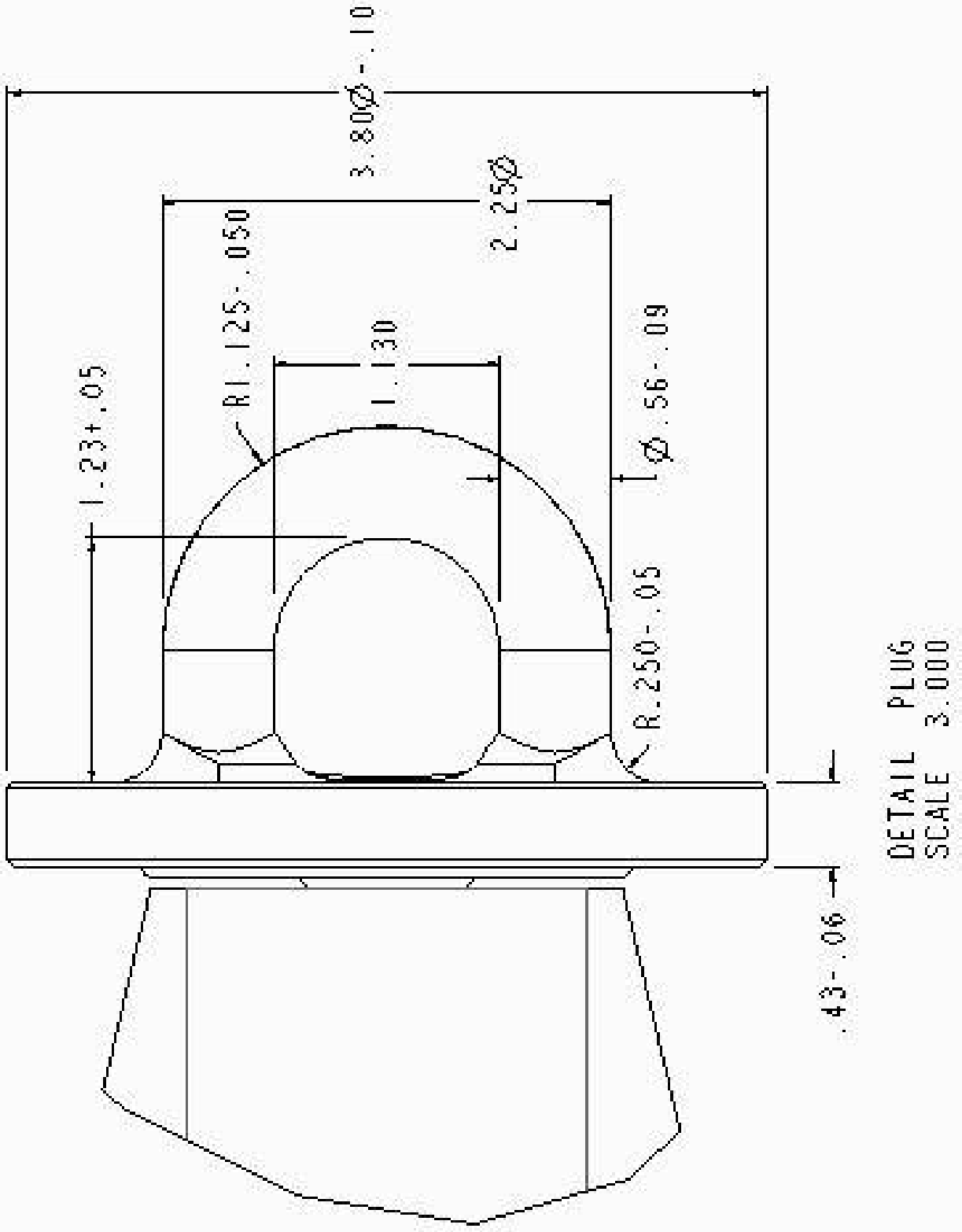


Figure 2. M107 Plug Detailed Interface Drawing

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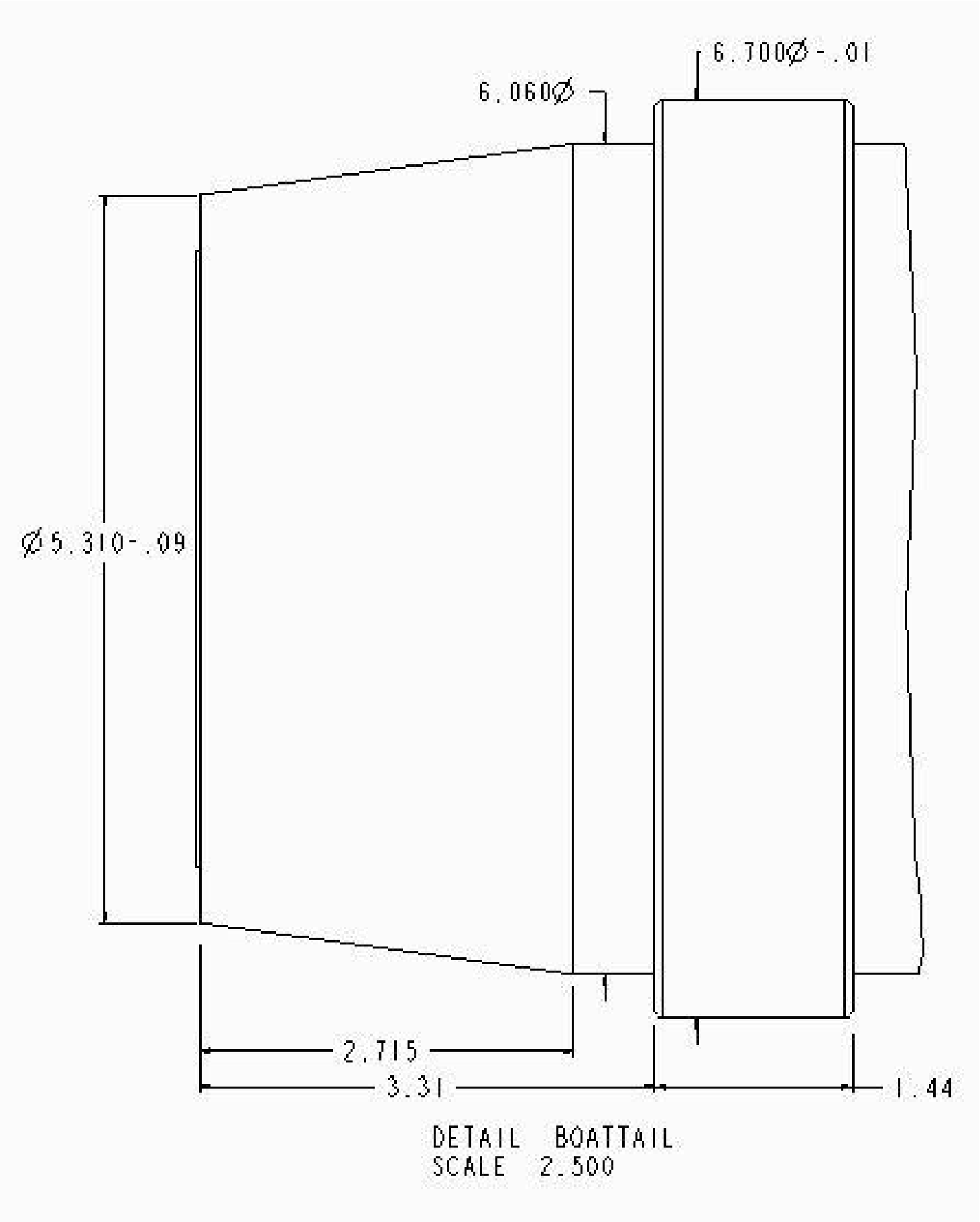


Figure 3. M107 Boattail Detailed Interface Drawing

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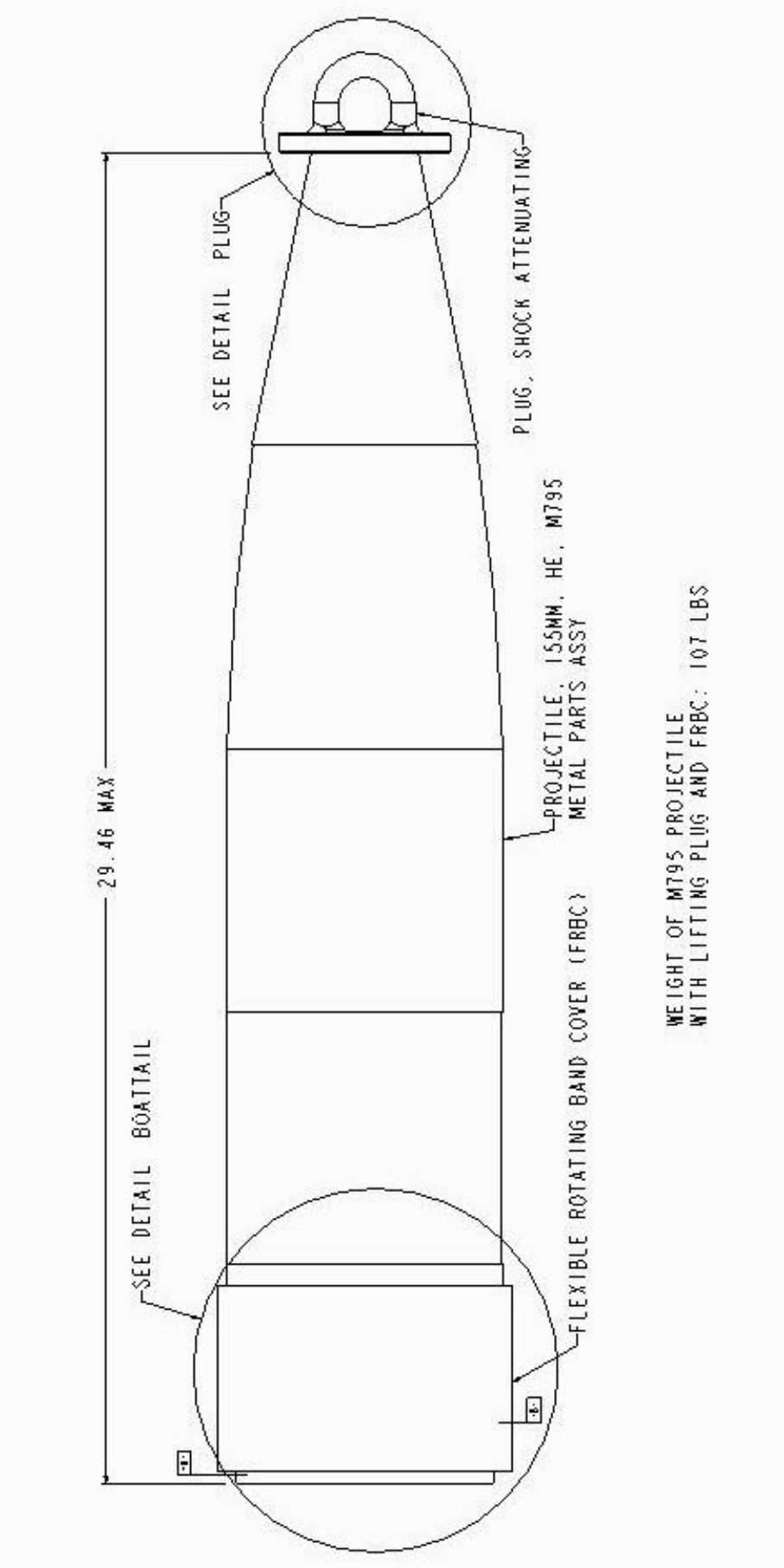


Figure 4. M795 Interface Drawing

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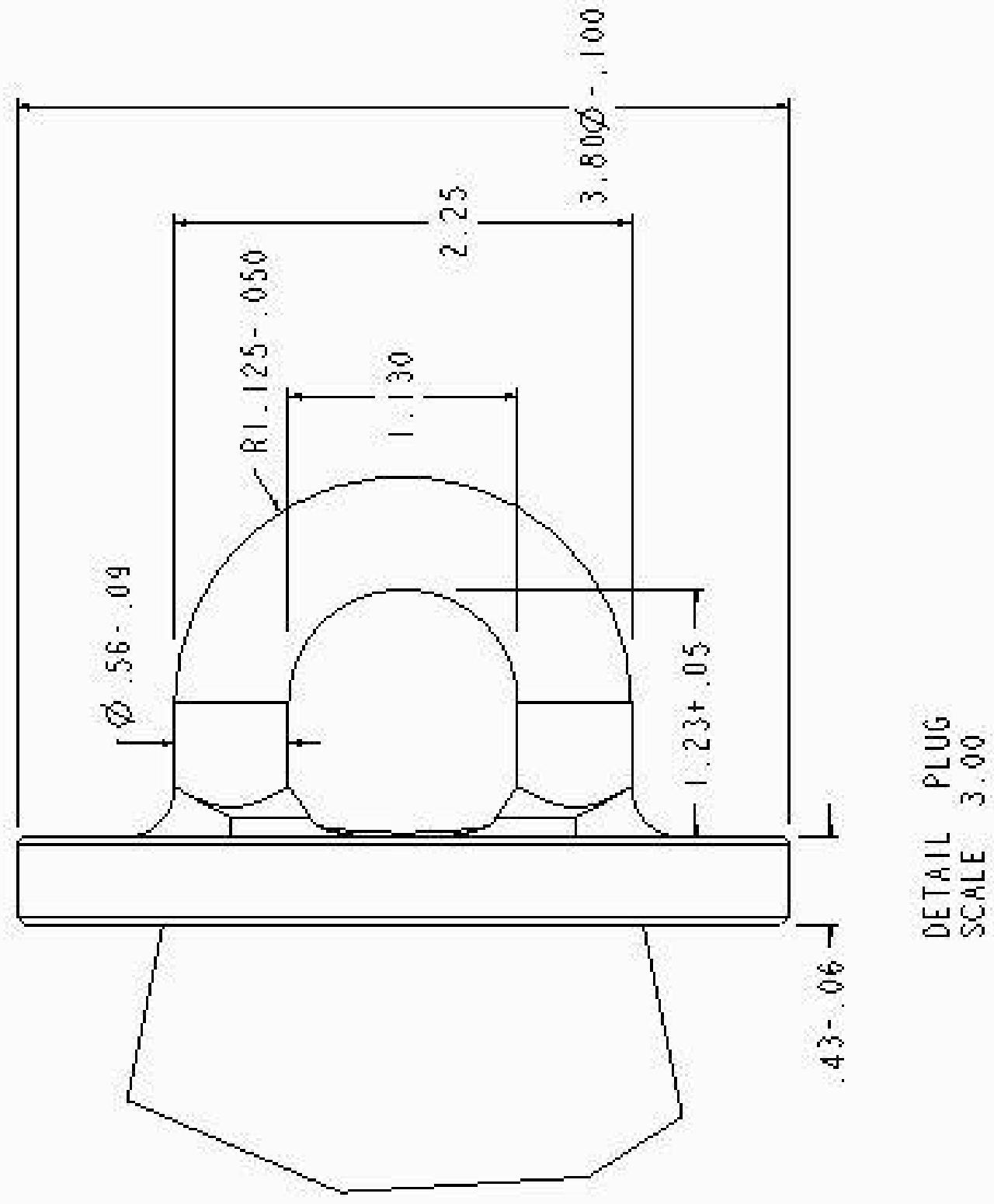


Figure 5. M795 Plug Detailed Interface Drawing

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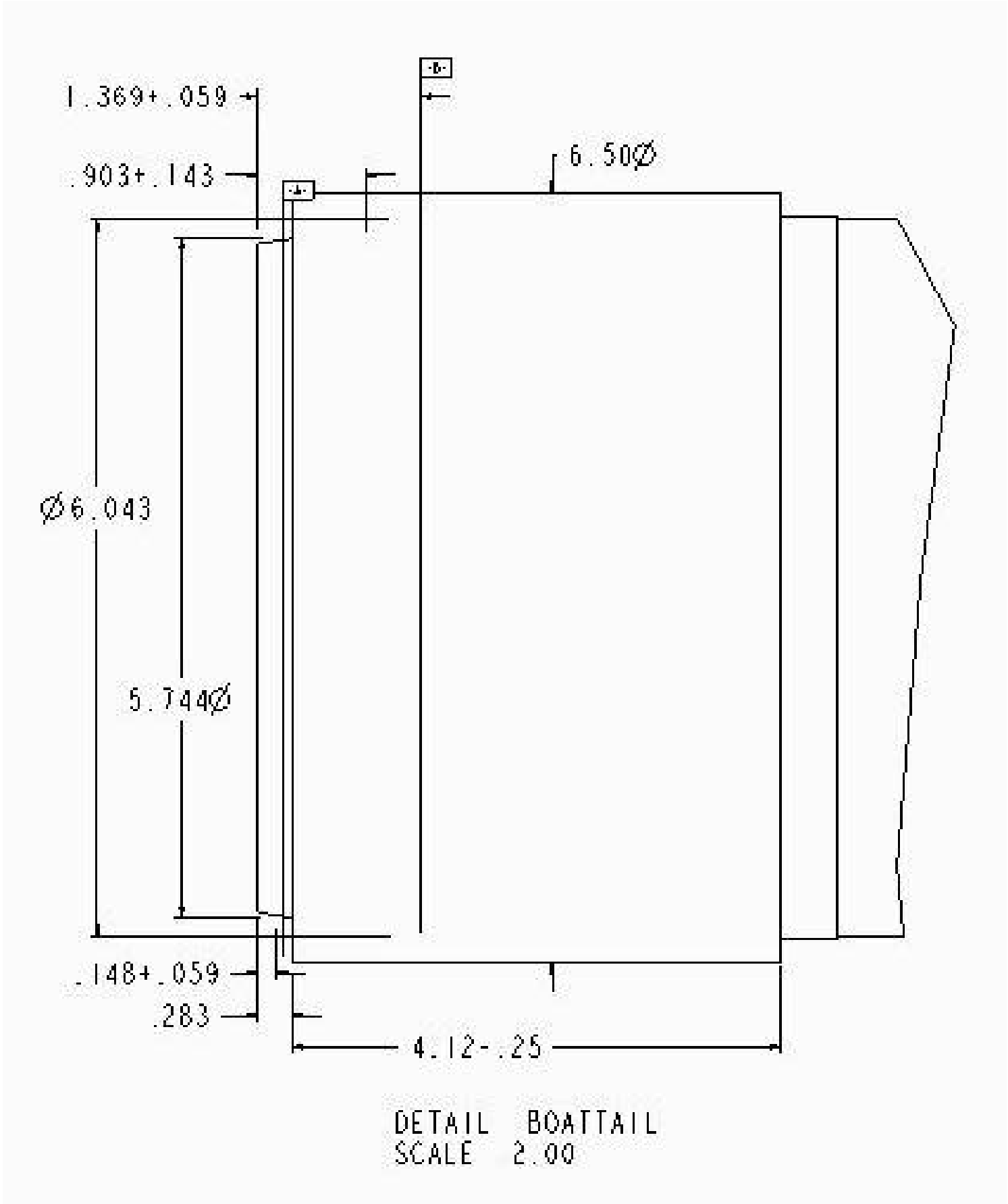


Figure 6. M795 Boattail Detailed Interface Drawing

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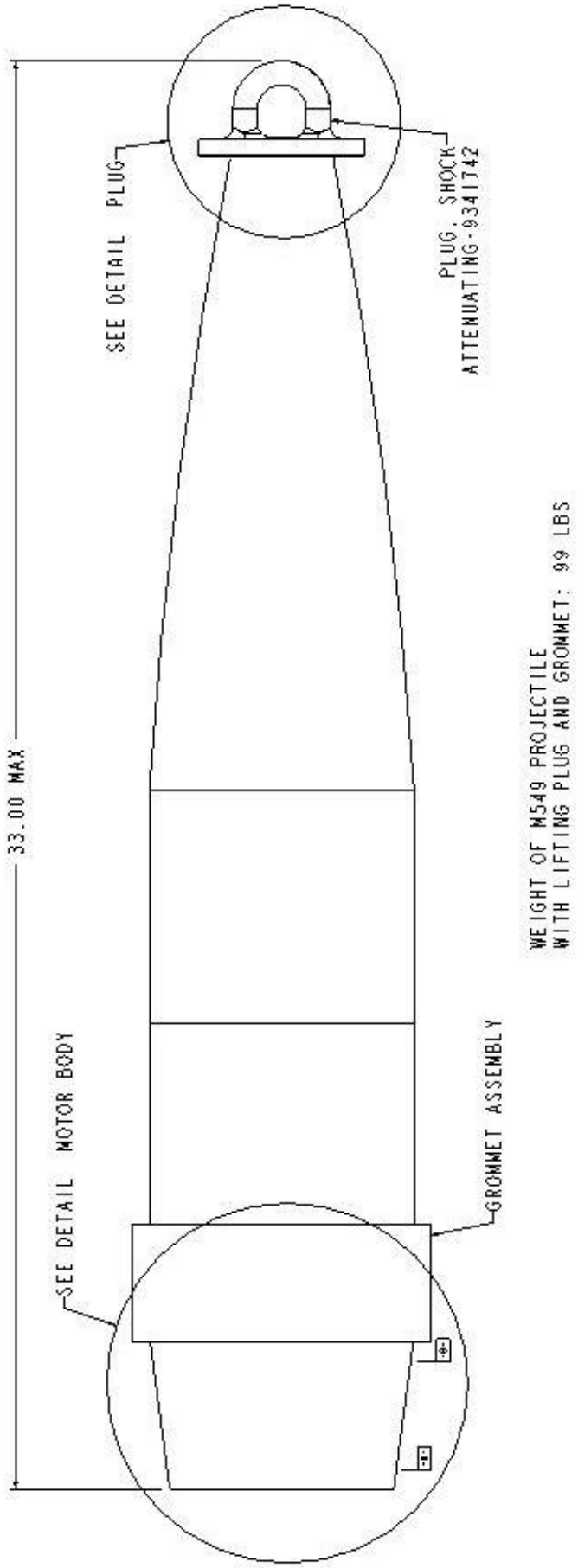


Figure 7. M549 Interface Drawing

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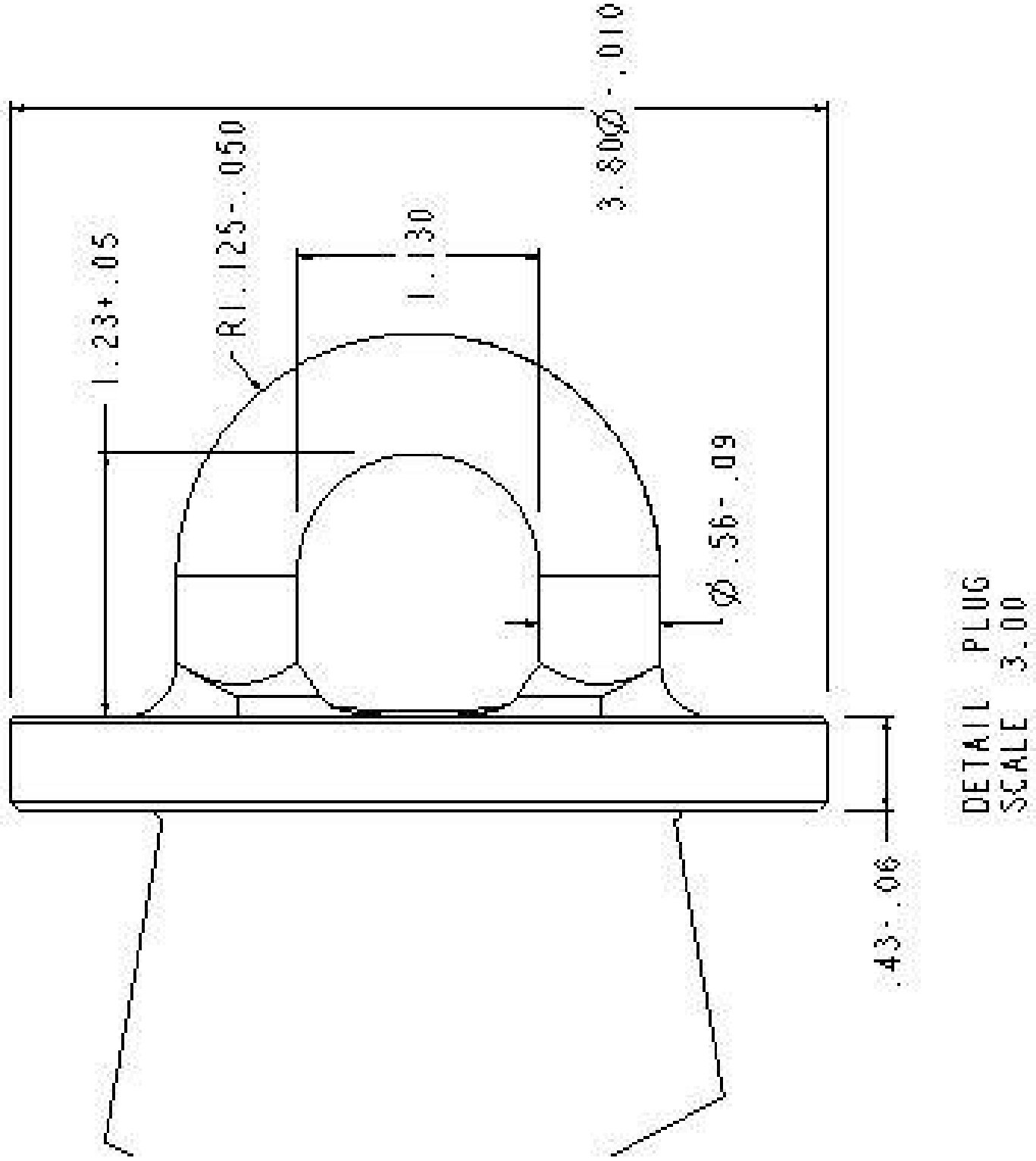


Figure 8. M549 Plug Detailed Interface Drawing

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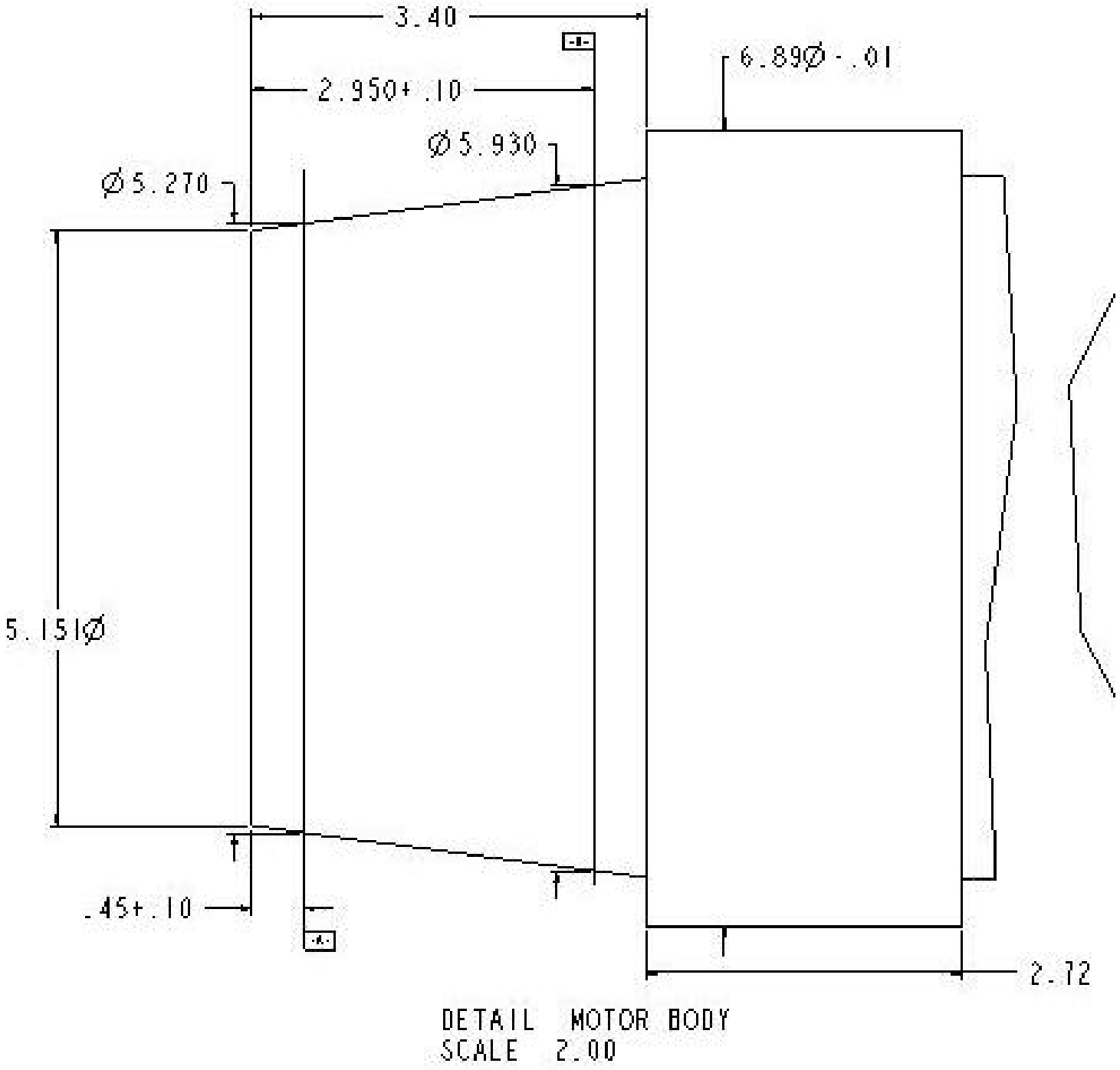


Figure 9. M549 Boattail Detailed Interface Drawing

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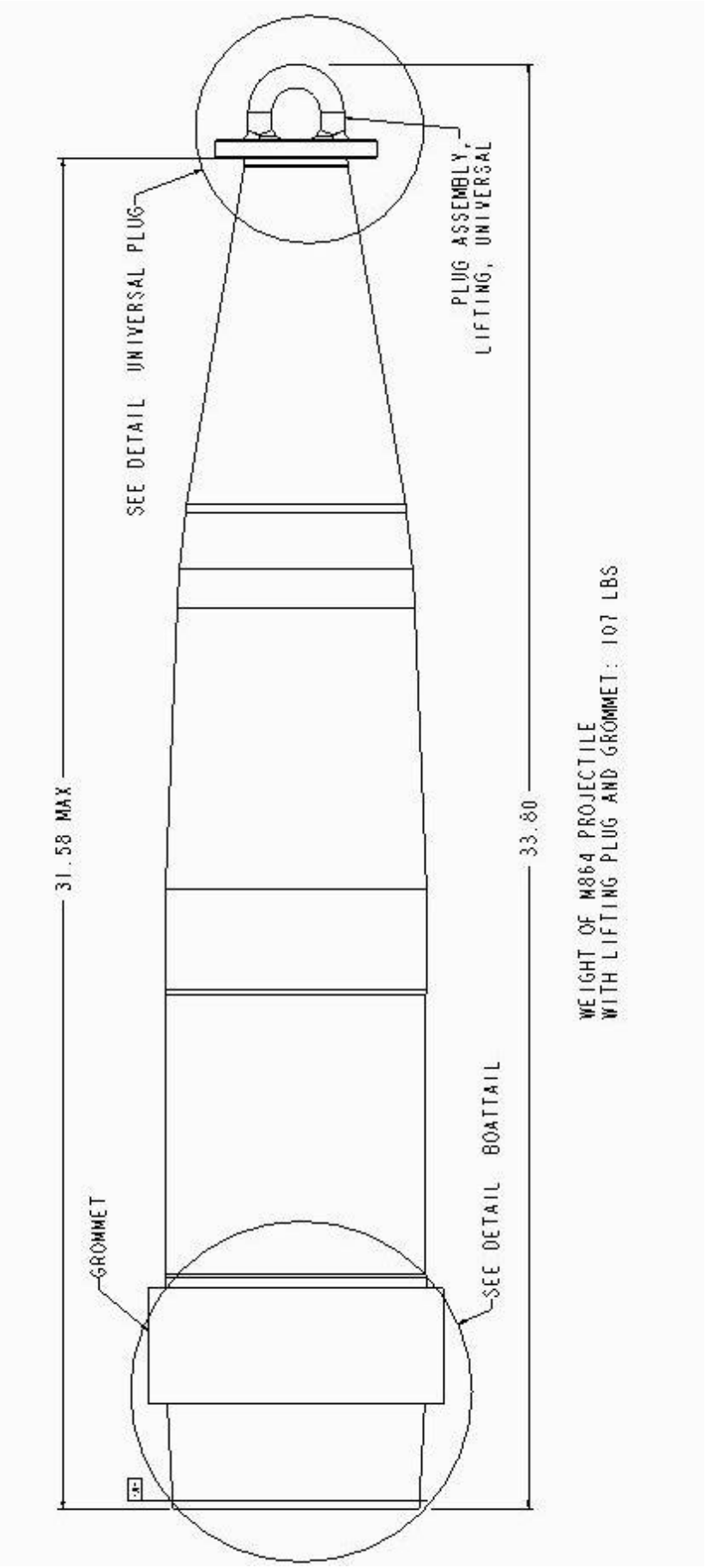


Figure 10. M864 Interface Drawing

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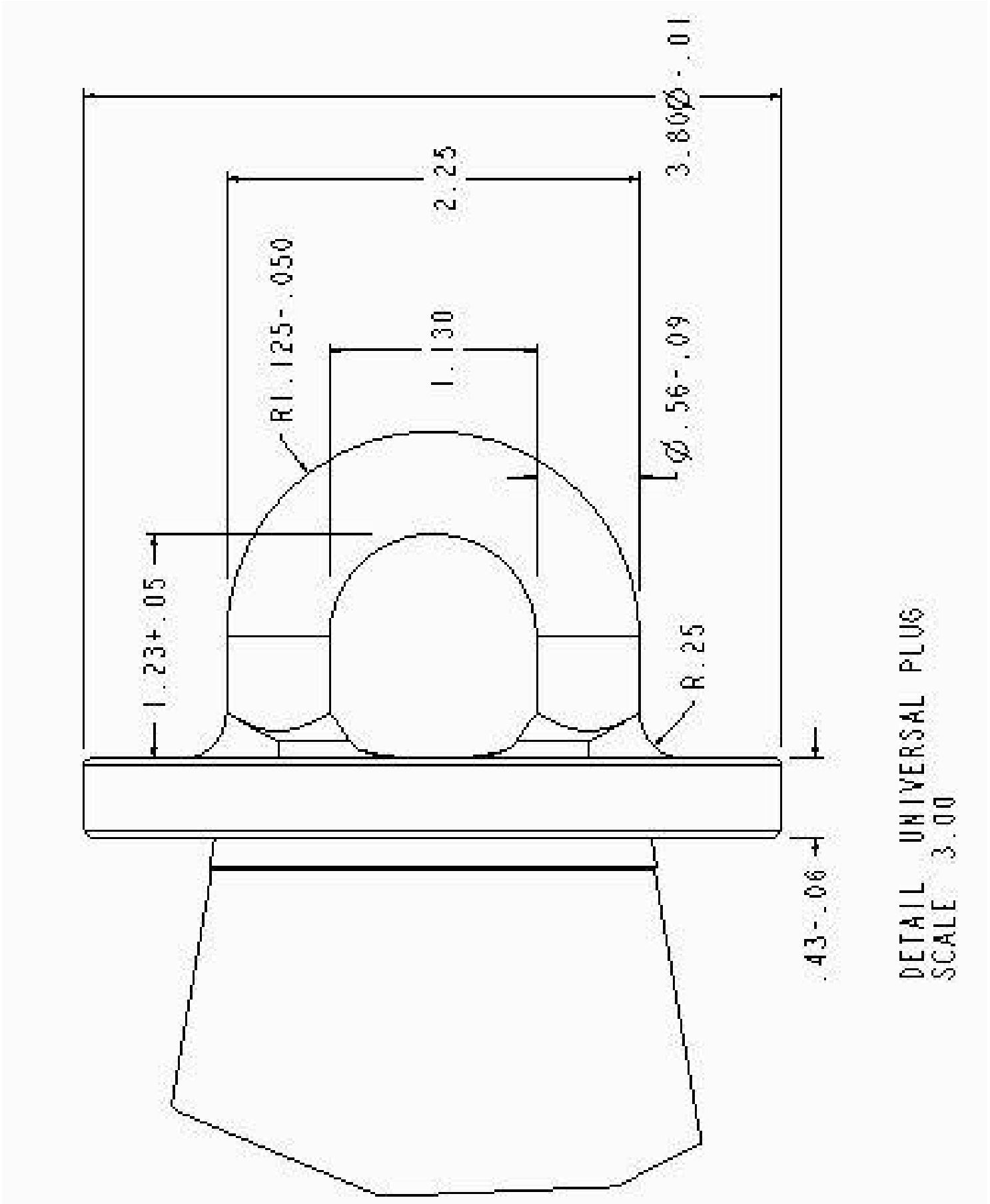


Figure 11. M864 Plug Detailed Interface Drawing

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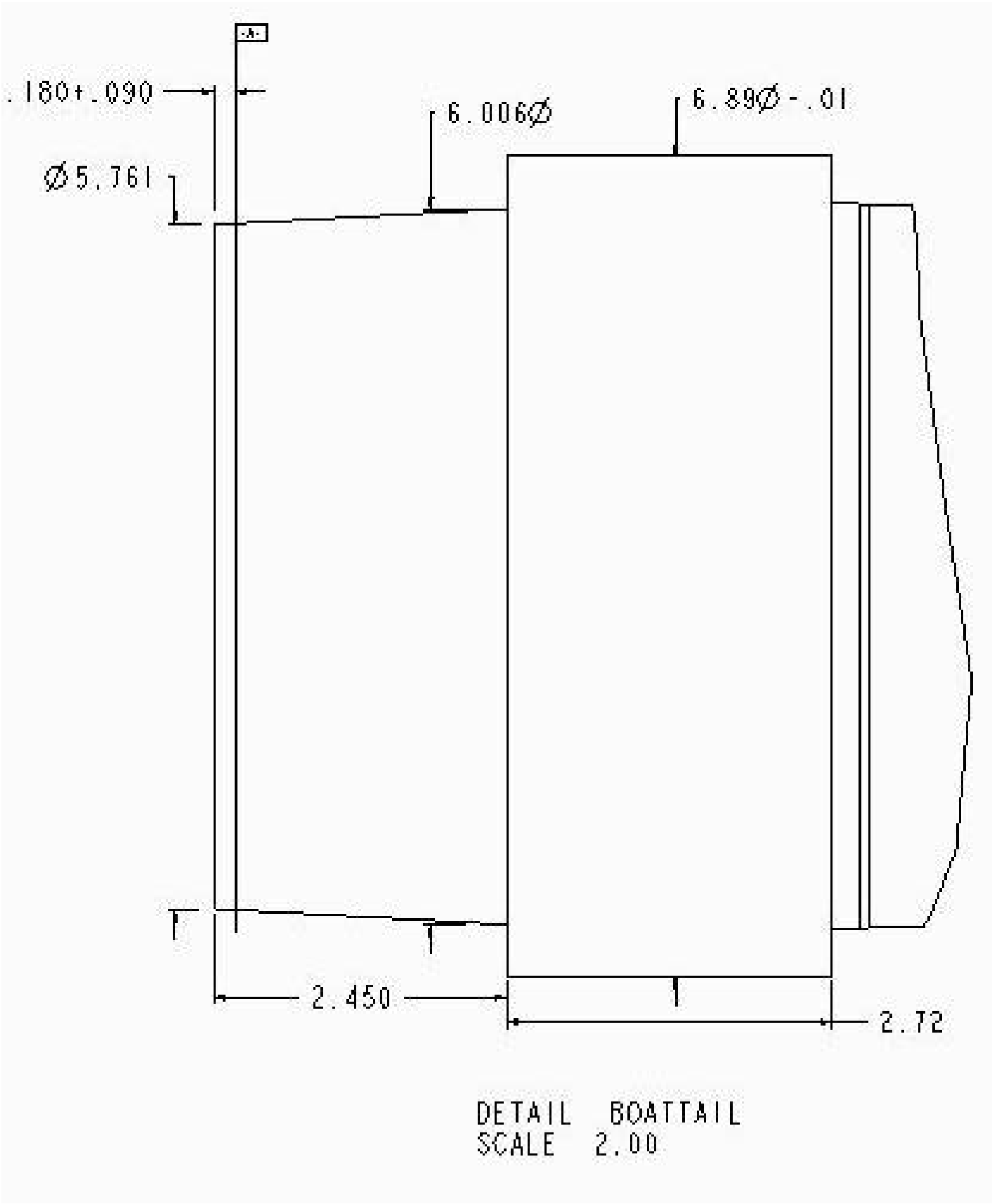


Figure 12. M864 Bottail Detailed Interface Drawing

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3.3.2. Ease of use. After projectiles have been placed on the pallet, any lock-down, closure, opening, or release actions shall be able to be accomplished by one person (Between the 95th percentile male and 5th percentile female). The pallet shall allow for access and removal of projectiles by grasping the lifting plug or associated area of the projectile. No tools shall be required to open and close TYPE I and TYPE III pallets. Common hand tools (such as strap cutters) are allowed to open and close TYPE II pallets. All pallets shall be designed to minimize the amount of loose pieces (maximum three) required to be removed for projectile access.

3.3.3 Color. All pallet surfaces shall be flat green in color. The shade for the interior and exterior surfaces will match the color chip for green, No. 34094 in accordance with FED-STD-595.

3.3.4 Chemical agent contamination survivability. The pallet shall be manufactured using materials which do not degrade after exposure to contaminants and decontaminants such that it would preclude mission accomplishment.

3.3.5 Identification. Each pallet shall have at least one area on the top with a minimum of 60 sq. inches which is clearly visible when the pallet is fully loaded, that can be painted or marked, and be capable of being marked over, using commercial paints or inks in accordance with ink, stencil, yellow No. 33533 Type I or IV of CID A-A-208 with no surface preparation other than cleaning and drying. Marking shall be clear and legible and shall remain on the pallet throughout its logistic life cycle.

3.4 Interface requirements. Each pallet shall accommodate the following inputs and interfaces.

3.4.1 Weight. The pallet, without projectiles, shall weigh no more than 70 pounds.

3.4.2 Dimensions. The assembled pallet shall be no more than 14.5 inches wide, 29 inches long, and the following maximum heights:

Table II Maximum heights

Projectile Family	Projectile Approximate Length	Maximum Heights
Style A - M107	23.8 inches	32 inches
Style B – M795	29.5 inches	37 inches
Style C – M549	30.6 inches	38 inches
Style D - M864	31.6 inches	39 inches

3.4.3 Transportation interfaces: The pallet shall be compatible with the following transportation modes and interfaces:

3.4.3.1 Forklifts. The pallet shall be compatible with being lifted and transported by a standard forklift, using approaches from either of two opposing longitudinal sides of the pallet. The pallet base shall have compatible interface dimensions for the purpose of being positioned on forklifts with existing pallets as shown in Figure 13.

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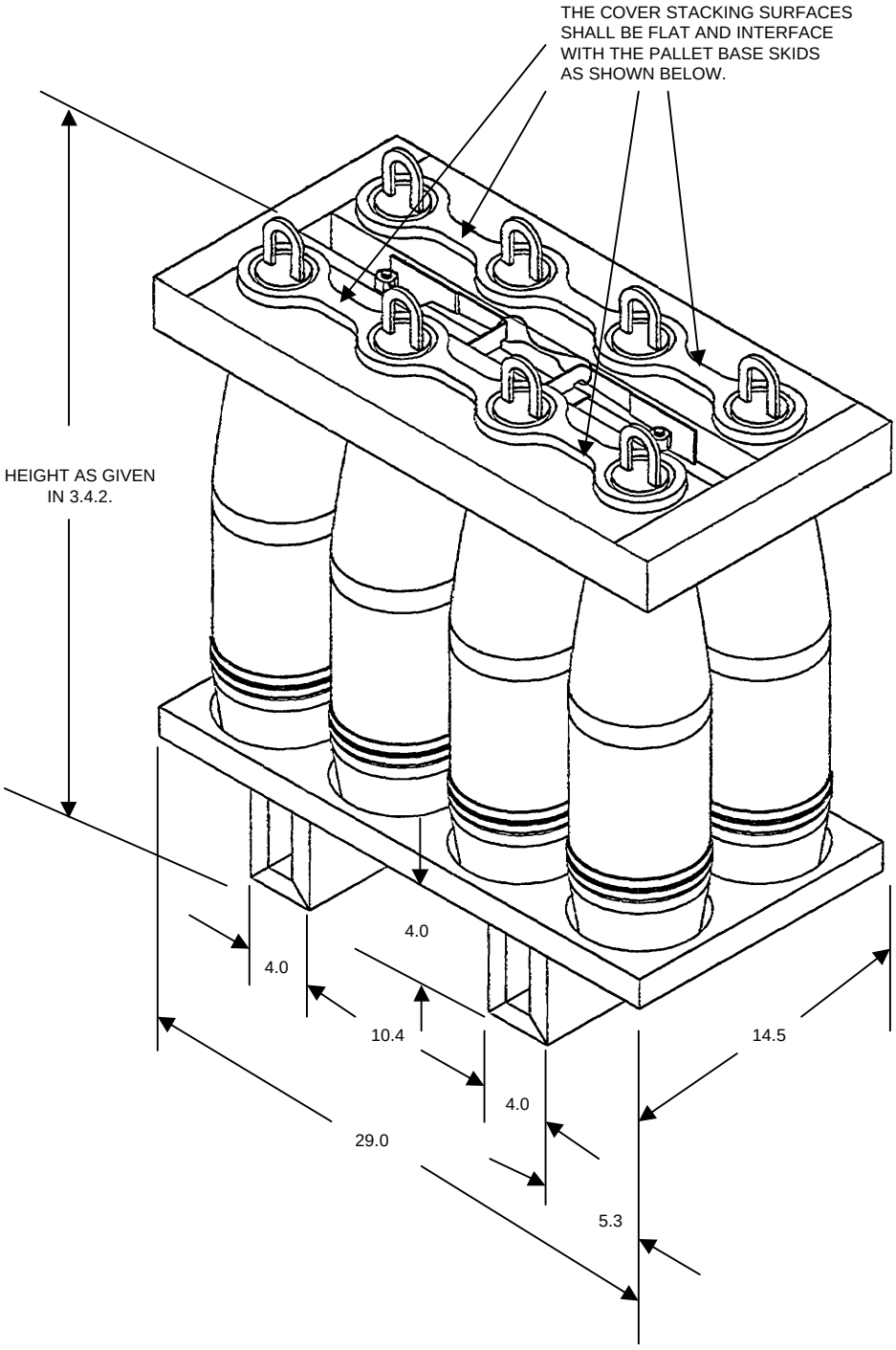


Figure 13. Pallet forklift and stacking interface dimensions

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3.4.3.2 Hoisting. The pallet shall have provisions for, and be capable of, being lifted by a four point sling through four lifting plugs and be capable of being lifted by one of the projectile lifting plugs.

3.4.4. Clothing and personal gear compatibility. The pallet and projectiles shall be capable of being loaded and unloaded by personnel wearing Arctic or Mission Oriented Protective Posture (MOPP) Level IV gear.

3.4.5 Unitization requirements. The pallet shall be capable of being strapped and bundled, sides flush together horizontally, with up to three (3) separate pallets, as shown in Drawing 19-48-4076. The pallet shall also be capable of being strapped and bundled with any combination of existing pallets of the same height. The bundled pallets shall remain stable while being transported and handled.

3.4.6 Interchangeability. The pallets of the same manufacturer, type and design shall functionally and physically interchange at its component level interfaces with other pallets of the same manufacturer, type and design.

3.4.7 Tamper evidence. The Type I and III pallets shall accommodate a standard wire security seal (drawing 8794342) so as to provide evidence of pallet opening by causing the seal to break.

3.5 Ownership and support requirements. The pallet shall possess the following life cycle ownership characteristics.

3.5.1 Design life. The pallet shall have a functional design life of 20 years of indoor storage (minimum exposure to rain, snow, or sunlight) and two years of outdoor storage without degradation to pallet and projectile.

3.5.2 Durability. The pallet shall continue to function properly through 100 open and close actions.

3.5.3 Safety. Each pallet shall present no physical handling hazards to the persons loading or unloading the pallet.

3.5.4 Pallet burning characteristics. The pallet and all associated components, except supporting dunnage, shall be non-combustible.

3.5.5 Reuse/disposal. The pallet shall be reusable. If damaged beyond reuse, the pallet shall be recyclable or easily disposed of as a non-hazardous material in a landfill.

3.5.6 Recycled, recovered, or environmentally preferable materials. Use of recycled, recovered, or environmentally preferable materials is desirable to the maximum extent possible provided the material meets or exceeds all specified requirements and promotes economically advantageous life cycle costs.

3.6 Environmental requirements. Each pallet shall operate under and after exposure to the following environmental conditions without damage or loss of performance. (except for forty foot drop)

3.6.1 Temperature. At low temperatures down to -65^{0}F. At high temperatures up to $+160^{0}$F.

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3.6.2 Stacking. Pallets shall be capable of nesting vertically, limiting horizontal sliding or shifting, and allowing for stacking up to four loaded pallets. Stacked pallets shall not bear on lifting plugs or projectiles on adjacent pallets. Pallets shall be compatible with stacking on or under previously designed pallets. Positioning of interface surfaces on the existing pallet covers and bases shall be as shown in the Figure 13.

3.6.3 Secured cargo vibration. After exposure to field service vibrations simulating transport in a "tie-down" condition by rail, truck, ship and tactical vehicle.

3.6.4 Drops. After severe drops onto steel surfaces that could occur during handling, maintenance, or issue operations.

3.6.5 Pendulum impact. After side impacts into steel surfaces that could occur while hoisting during ship loading/unloading operations.

3.6.6 Loose cargo vibration. After exposure to field service vibrations encountered during transport over rough roads in an unrestrained condition.

3.6.7 Forty foot drop. The pallet shall prevent the projectiles from detonating and allow the projectiles to be safely disposed of, after exposure to drop conditions such as accidental free fall of packaged ammunition during ship loading/unloading operations. The pallet does not have to function after this test.

3.6.8 Railcar impact. After exposure to the impact environment found during rail transportation.

3.6.9 Temperature shock. The pallet shall withstand a sudden change in ambient temperature.

3.6.10 Salt spray. The pallet shall be safe to handle and be operable during and after a minimum of 336 hours salt spray fog exposure without evidence of corrosion.

3.6.11 Chemicals, petroleum, oil, and lubricants. After exposure to chemical, petroleum, oil, and lubricant based products.

3.6.12 Sand. While exposed to blowing sand.

3.6.13 Dust. While exposed to dusty atmosphere.

3.6.14 Solar (UV) radiation. After exposure to direct solar (UV) radiation during unsheltered storage on the earth’s surface.

3.6.15 Fungus. After exposure to tropical climate fungus, rot, or mildew.

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4. VERIFICATION

TABLE III. REQUIREMENT/VERIFICATION CROSS REFERENCE MATRIX

METHOD OF VERIFICATION	CLASSES OF VERIFICATION	LEVELS	LEVELS
N/A – NOT AVAILABLE 1 – ANALYSIS 2 – DEMONSTRATION 3 – EXAMINATION 4 – TEST	A = FIRST ARTICLE B = CONFORMANCE	MIL-STD-1916	

SECTION 3 REQUIREMENT	VERIFICATION METHOD					VERIFICATION CLASS		SECTION 4 VERIFICATION	VL
	N/A	1	2	3	4	A	B		
3.3	X							4.3	
3.3.1			X			X		4.3.1	
3.3.1.1			X			X		4.3.1.1	
3.3.2			X			X		4.3.2	
3.3.3				X		X	X	4.3.3	II
3.3.4					X	X		4.3.4.1-4.3.4.3	
3.3.5				X		X	X	4.3.5	III
3.4	X							4.4	
3.4.1					X	X	X	4.4.1	II
3.4.2					X	X	X	4.4.2	II
3.4.3	X							4.4.3	
3.4.3.1					X	X	X	4.4.3.1	I
3.4.3.2					X	X	X	4.4.3.2, 4.4.3.2.1	I
3.4.4			X			X		4.4.4	
3.4.5			X				X	4.4.5	I
3.4.6			X			X		4.4.6	
3.4.7					X	X	X	4.4.7	I
3.5	X							4.5	
3.5.1					X	X		4.5.1	
3.5.2					X	X		4.5.2	III
3.5.3				X		X	X	4.5.3	III
3.5.4					X	X		4.5.4	
3.5.5					X	X		4.5.5	
3.5.6				X		X		4.5.6	
3.6	X							4.6	
3.6.1					X	X		4.6.1	
3.6.2					X	X		4.6.2	
3.6.3					X	X		4.6.3	
3.6.4					X	X		4.6.4.1-4.6.4.3	
3.6.5					X	X		4.6.5	
3.6.6					X	X		4.6.6	
3.6.7					X	X		4.6.7	
3.6.8					X	X		4.6.8	
3.6.9					X	X		4.6.9	
3.6.10					X	X		4.6.10	
3.6.11					X	X		4.6.11.1-4.6.11.2	
3.6.12					X	X		4.6.12	
3.6.13					X	X		4.6.13	
3.6.14					X	X		4.6.14	
3.6.15					X	X		4.6.15	

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4.1 Classification of verifications. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1).
b. Conformance inspections (see 4.1.2).

4.1.1 Verification conditions. Unless otherwise specified, all inspections shall be performed on all types of pallets in accordance with the test conditions specified in the paragraphs in Table III.

4.1.1.1 First article verification. First article inspection shall be performed with a maximum of thirty nine (39) production-representative units of an order when a first article sample is required (see 3.1). This inspection shall include the verifications listed in Table III. The following Table IV shows approximately the number of pallets needed for first article inspection.

4.1.1.2 First article rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected.

Table IV Number of pallets for first article

Number of Pallets	Type of Test
6	4.3.1, 4.3.1.1, 4.3.2, 4.3.3, 4.3.5, 4.4.1, 4.4.2, 4.4.3.1, 4.4.3.2, 4.4.3.2.1, 4.4.4, 4.4.6, 4.4.7, 4.5.2, 4.5.3, 4.5.5, 4.5.6, 4.6.9, 4.6.11
1	4.3.4
3	4.4.5
1	4.5.1
1	4.5.4
6	4.6, 4.6.1, 4.6.2, 4.6.3, 4.6.4, 4.6.4.1, 4.6.4.2, 4.6.4.3, 4.6.5, 4.6.6
3	4.6.7
12	4.6.8
2	4.6.10
1	4.6.12
1	4.6.13
1	4.6.14
1	4.6.15
39	

4.1.1.3 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of Table III. The first article sample shall consist of the assemblies, components and test specimens listed above in the quantities indicated.

4.1.2 Conformance inspection. Conformance inspection shall include those examinations and tests from Table III as defined in the contract. Conformance testing shall be performed on all types of pallets with seven (7) pallets as shown in the following Table V:

Table V Number of pallets for conformance inspection

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Number of Pallets	Type of Test
6	4.4.1, 4.4.2, 4.4.3.1, 4.4.3.2, 4.4.3.2.1, 4.4.5
1	4.3.3, 4.3.5, 4.4.7, 4.5.3

7	

4.1.2.1 Lot formation. Lot formation shall be in accordance with the lot formation requirement of MIL-STD-1916, paragraph 4.2.

4.1.2.2 Acceptance criteria. If any component, assembly or test specimen fails to comply with the applicable requirements, the lot shall be rejected.

4.2 Verification methods. The types of verification methods included in this section are visual inspection, measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously-approved or previously-qualified designs. See 6.8 for recommended test facilities.

4.2.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment in a plan approved by the government, including the application of statistical process control, tool control, or cost-effective sampling procedures, to verify performance.

4.2.2 Order of inspection. The inspection sequence is at the discretion of the contractor. Combining tests for efficiency is allowable if the desired verification results for each requirement remain distinguishable.

4.2.3 Sequential packed rough handling test. The purpose of these tests is to demonstrate that the contractor's design will provide adequate protection to the 155mm projectiles such that they will continue to function reliably having experienced these environments. Six pallet assemblies containing 8 projectiles each, will be tested; two at ambient temp, two at $+160^0$F and two at -65^0F. Each pallet shall be operable during and after the environmental conditions listed in Figure 14 without opening or losing its contents and must remain functional and suitable for its intended purpose without damage or loss of performance. It is considered functional if it meets the following criteria:

a. The pallet cover is capable of being unlatched, opened, re-closed, and latched shut.

b. Stacking or handling capability is not impaired.

c. Projectile support and protection is maintained.

d. Projectiles can be removed from and repacked onto the pallets as intended.

e. Projectiles shall not be damaged in any manner. There shall be no degradation of the projectile finish and no material adhering to the projectile.

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Sequential Packed Rough Handling Test Sequence
STACKING TEST
SECURED CARGO VIBRATION TEST
CORNER, EDGEWISE, ROLLOVER DROP TESTS
LOOSE CARGO VIBRATION TEST

Figure 14 - Transportation vibration handling test

Note: Upon successful completion of all the Sequential Packed Rough Handling test, the next series of material tests can be conducted on empty, new or acceptable used pallets depending upon their physical condition.

4.3 Operating requirements verification. Complete each verification under paragraph 4.3.

4.3.1 Projectiles storage inspection. Demonstrate that the TYPE I and III pallets can vertically store two to eight 155MM projectiles. Demonstrate that the projectile cannot be moved by hand laterally more than .25 inches, when locked in the pallet. Visually verify that the TYPE II pallet can vertically store eight 155MM projectiles. Demonstrate that the TYPE II pallet can be strapped with 1 ½ inch strapping. Demonstrate that the TYPE III pallet can vertically store all lengths of projectiles in 3.3.1.1. Demonstrate that a grommet or flexible rotating band cover can be attached and removed. Examine that the projectiles are held at the lifting plug and boattail. Visually verify and review the pallet drawings that the materials used to hold the boattail will not cause metal to metal abrasion or corrosion. Demonstrate that the pallet can be disassembled to minimize space.

4.3.1.1 Projectile length verification. Demonstrate that the pallet can store projectiles of the length specified in the contract by loading a pallet with projectiles and closing.

4.3.2 Ease of use verification. Six (6) pallets shall be packed with dummy projectiles. All pallets shall be closed and security seals installed at the start of the test. The pallets shall be able to be opened and closed by one person exerting a force of 20 +/-5 pounds. The projectiles shall be able to be removed without the use of special tools. For TYPE I and III pallets, the pallet shall be opened and closed without the use of special tools, excessive force or damage to the pallet. (See 6.11) For TYPE II pallets, no security seal is required. TYPE II pallets shall be opened and closed with the use of common tools but with no excessive force or damage to the pallet. Demonstrate access to the first projectile within 15 seconds or less 90 percent of the time. Each additional projectile shall be able to be removed within another 10 seconds.

4.3.3 Color verification. The interior and exterior surfaces of the pallet will match the color chip for green, No. 34094 in accordance with FED-STD-595.

4.3.4. Chemical agent contamination survivability verification. Compliance shall be demonstrated by testing the external material of the pallet in accordance with Test Operations Procedure (TOP) 8-2-

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111. After the tests, the pallet shall still meet the ease of use verification found in 4.3.2. Also, after the test, the pallet shall meet the test criteria of 4.3.4.1, 4.3.4.2, and 4.3.4.3. See 6.8.f for recommended test facility.

4.3.4.1 Decontamination. After complete decontamination washing and drying, verify the pallet still meets the test criteria of 4.3.2.

4.3.4.2 Residual vapor hazard (calculated for outdoor exposure scenario) This translates to post-decon agent monitoring for 12 hours. The maximum allowable dosages (cumulative) for human threshold are as follows:

HD - 50 mg min/m3
VX - .25 mg min/m3
TGD - 2.5 mg min/m3

Calculate the maximum allowable desorption from the item (pallet/ package) using the following equation:

$$Q = D \times \pi \times u \times Gy \times Gz$$

Where:

Q = maximum allowable desorption (12 hour cumulative, mg)
D = maximum allowable dosages (mg min/m3)

$\pi = 3.14$

u = wind speed (m/min) Note: Maximum worst case as per HQ DA is 60 m/min or 1 m/sec.

Gy = (Gamma y) = item vertical dimension divided by 4.3 (m)
Gz = (Gamma z) = item crosswind dimension divided by 4.3 (m)

Using the values listed above to perform the calculations, the maximum allowable desorptions can be derived for an outdoor exposure scenario. Compatibility and hardness must be within default limits of 15 and 20 percent degradation respectively

4.3.4.3 Residual contact hazard. In addition to measuring residual vapor hazards, the pallet shall be subjected to testing to determine its residual contact hazard. Residual contact hazard as measured over a 12 hour period (cumulative) for an exposed skin surface area of 1200 cm2 (based on a 70 kg male), shall not exceed:

Agent	mg/70kg male
VX	1.4
HD	180.0
GD	30.0

4.3.5. Identification. Verify by demonstrating that the pallet will accept standard commercial ink, stencil, yellow No. 33533 Type I or IV of CID A-A-208 and remain legible after rough handling testing.

4.4 Interface requirements verification. Complete each verification under paragraph 4.4.

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4.4.1 Weight inspection. Demonstrate by weighing that each of the six pallet assemblies without projectiles, weighs less than 70 pounds.

4.4.2 Pallet dimensions verification. The external dimensions of six (6) pallets shall be measured. The maximum external dimensions when assembled with 8 projectiles, shall not exceed 14.5 inches in width, 29 inches in length, and the heights specified in 3.4.2 for each of the pallets.

4.4.3 Transportation interfaces verifications: Complete each of the following transportation interface verifications with a fully loaded pallet of the heaviest projectiles for TYPE II and III pallets. TYPE I pallets shall be tested with the heaviest projectile of the design length being procured. (see 6.2)

4.4.3.1. Forklift truck compatibility. Perform this test in accordance with Method 5011 of FED-STD-101. A fully loaded pallet is raised 6 inches off the floor with a forklift truck. The forklift truck is driven over 3 sets of 1 inch thick pieces of wood spaced 30 feet apart on 100 foot course. The wood is positioned at angles of 90 degrees, 60 degrees and 75 degrees to the forklift tires. During the test the pallet shall remain stable on the forklift tongs. After the test, the pallets shall show no stress cracks, and no deformations. Measure the forklift skids or any protrusions on the bottom of the pallet, intended to interface forklifts, and show by comparison to comply with the spacing shown in Figure 13.

4.4.3.2. Hoisting. The loaded pallet shall be loaded to four times its gross (loaded) weight and shall be lifted free of the ground by means of the lifting provisions provided. The pallet shall be maintained in the position for two minutes. After the test, the pallets must show continued ease of opening per 4.3.2, no stress cracks, and no major deformations.

4.4.3.2.1 Sudden lift. Two diagonally opposite projectile lifting plugs on the fully loaded pallet shall be engaged and lifted by a short two-legged sling. The load shall be allowed to free fall 4 inches and brought to an abrupt stop. This shall be done two more times through only a single lifting plug located diagonally opposite one another. After the test, the pallets must show continued ease of opening per 4.3.2, no stress cracks, and no deformations.

4.4.4. Clothing and personal gear compatibility demonstration. With an operator wearing, in turn, a full standard MOPP-IV uniform and then full standard arctic uniform, demonstrate ease of use per 4.3.2 while the operator is in each uniformed configuration.

4.4.5 Unitized stress test. The following tests are conducted on a unitized set of three pallets loaded with 8 projectiles each. After each test is performed the pallets shall show continued ease of opening per 4.3.2, no damage to the projectiles, no stress cracks, only minor deformations, and the pallets shall remain unitized. TYPE I pallets shall be tested with the heaviest projectile of the design length being procured. TYPE II and III pallets shall be tested with heaviest and longest length projectile.

a. Stacking test. The assembled unit load is subjected to a load of 10,000 pounds stacked on top for one hour. This weight is equivalent of four unit loads high.

b. Repetitive shock test. This test shall be conducted in accordance with Method 5019 of FED-STD-101. The assembled unit load is placed on the vibration table and vibrated at a frequency of 3.4 Hz for two hours.

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c. Edgewise drop test. This test shall be conducted in accordance with Method 5008 of FED-STD-101. One edge of the unit is placed on six inch block. The opposite side is raised to a height of 15 inches and allowed to free fall. The edge is impacted onto an unyielding concrete surface. All four edges are to be dropped.

d. Cornerwise drop test. This test shall be conducted in accordance with Method 5005 of FED-STD-101. Two adjacent corners of the load are raised to a height of 6 and 12 inches. The corner opposite the 12 inch block is raised to a height of 18 inches and allowed to free fall onto an unyielding surface. All four corners are to be impacted.

e. Impact test. This test shall be conducted in accordance with Method 5023 of FED-STD-101. The unit load is placed on the carriage of the inclined impact tester. The carriage is raised to a predetermined point where the impact velocity is 7 feet per second. All four sides are to be impacted.

f. Tipover test. This test shall be conducted in accordance with Method 5018 of FED-STD-101. The unit is placed on a tilt platform and raised to form a 20 degree angle. This test is repeated on an adjacent side of the load.

g. Forklift truck compatibility. This test shall be conducted in accordance with Method 5011 of FED-STD-101. The unit load is raised 6 inches off the floor with a forklift truck. The forklift truck is driven over 3 sets of 1 inch thick pieces of wood spaced 30 feet apart on 100 foot course. The wood is positioned at angles of 90 degrees, 60 degrees and 75 degrees to the forklift tires.

h. Sling compatibility. The pallet load is lifted, swung, lowered and handled with sling. The load is also pulled back 9 inches above level and released into a concrete wall to see if the sling would disengage.

i. Unstacking Test. The pallet load is placed on top of another pallet load and then removed with a forklift.

4.4.6 Interchangeability demonstration. Perform an interchangeability test on one manufacturer, type and design by disassembling and staggering the distribution of components specified in the interchangeability requirement into five equal sets of parts. Reassemble each set of parts and mate them to the remaining components so that five different complete pallets result. Perform an opening and closing demonstration on each pallet and demonstrate ease of use in accordance with 4.3.2.

4.4.7 Tamper evidence demonstration. Verify by opening the pallet cover that the seal breaks giving evidence of opening. Verify that force or separate operation is needed to break the seal. Verify that a projectile can not be removed without breaking the seal.

4.5 Ownership and support requirements verification. Complete each verification under paragraph 4.5.

4.5.1 Design life evaluation. (Accelerated heat aging/ environmental stress cracking evaluation) - One (1) empty pallet shall be used for the test. Stabilize oven/ chamber to 160^0F. Examine the condition of the pallet. Examine every nick or possible crack. Apply 100 percent pure reagent made of a surfactant such as a concentrated soap solution to the pallet by either dipping or brushing. For components apply 100 percent pure reagent throughout the part. Cover all pallet/ components/ materials with a

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polyethylene covering (bags) to prevent evaporation. Place pallet/ components/ materials in the oven/ chamber and observe time, date and oven/ chamber temperature. Examine the pallet/ components/ materials after the first four hours and observe any physical changes including hairline cracks using sketches and photographs. Inspect twice a day after the first 24 hours for a total of 120 hours. Reapply reagent to the pallet areas daily or more frequently as and when reagent evaporates. Observe the beginning of a hairline crack (s) to the development of a through crack (s). Observe time, date, and temperature of oven/ chamber when the observations are made. Acceptance criteria shall be no cracking greater than one inch in length and continued ease of opening per 4.3.2.

4.5.2 Durability evaluation. Verify that the pallet shall be capable of being repeatedly opened and closed, 100 times without loss of product protection capability. Six (6) pallets shall be packed with dummy projectiles in accordance with 4.3.2. All pallets shall be closed at the start of the test. Inspect the pallets for proper function.

4.5.3 Safety verification. Inspect the pallet to be free of hazardous burrs, cracks, deformation, distortion, disconnects, improper serration, rings, bulges, or sharp edges that would prevent proper seating alignment, engagement, or assembly of pallet components, posing physical danger to the operator or equipment.

4.5.4 Pallet burning characteristics. One (1) pallet each except for supporting dunnage shall be subjected to the flame spread index (ASTM E162), smoke density (ASTM E662), and toxic products of combustion (ASTM E800) tests. The acceptance criteria of each tests shall be as follows:

a. The pallet material shall have a flame spread index of 25 or less when tested in accordance with ASTM E162.

b. The pallet material shall have a smoke density of 200 or less when tested in accordance with ASTM E662.

c. The pallet material shall be free of all toxic chemicals when tested in accordance with ASTM E800. (See 6.11)

4.5.5 Reuse/disposal. Visually verify that the pallet can be reused after testing. Review material certifications that the pallet materials can be recycled or easily disposed of in a landfill.

4.5.6 Recycled, recovered, or environmentally preferable materials. Verify that recycled, recovered, or environmentally preferable materials is used to the maximum extent possible at an economically advantageous life cycle cost.

4.6 Operating environment requirements verification. For sequential rough handling testing (see 4.2.3), conduct ambient temperature ($+70^0$F +/- 20^0F) testing first in accordance with Figure 2 (4.6.1 - 4.6.5) prior to performing temperature testing. For sequential rough handling testing the projectiles may be inert or live rounds. The acceptance criteria for the pallet with inert and live rounds shall be per 4.3.2 with no stress cracks and only minor deformations. In addition, the projectiles shall be free from cracks, deformation, or other damage. For live rounds the projectile shall be fired and function properly following the applicable procedures in Appendix A or B (except for projectiles subjected to forty foot drop test 4.6.6). Verify visual that there is no degradation of the projectile finish or any material

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adhering to the projectile. See 6.8 for test facilities. See 6.9 for live or inert testing requirements and quantities of Government Furnished Material (GFM) live and inert projectiles.

4.6.1 Temperature evaluation. Condition two test pallets for at least twenty-four (24) hours at $+160^0$F. Conduct testing in accordance with 4.6.3 thru 4.6.5. Acceptance criteria shall be in accordance with 4.3.2 and 4.4.3. Acceptance testing may be performed at ambient temperature ($+70^0$F +/- 20^0F). Allow the temperature to stabilize at ambient $73^0 \pm 4^0$F for 24 hours. Condition the two test pallets for at least twenty-four (24) hours at -65^0F. Conduct testing in accordance with 4.6.3 through 4.6.5. Acceptance criteria shall be in accordance with 4.3.2.

4.6.2 Stacking test. Four fully loaded pallets shall be stacked one on top of another on the top of the loaded pallet for a minimum of 24 hours. All pallets shall remain stable with no evidence of sliding or shifting that would cause an unsafe condition. After the test, the pallets shall show continued ease of opening per 4.3.2, with no stress cracks, and no deformations. Verify the dimensions of the pallet base and pallet cover meet Figure 13. Visually verify that the pallets nest on top of each other correctly. The above pallet shall not bear upon the lifting plugs of the lower pallet.

4.6.3. Secured cargo vibration. The six (6) conditioned (for 24 hours minimum) test pallets (2 hot, 2 cold, 2 ambient) shall be subjected to the secured cargo vibration test. The test shall be conducted in accordance with 4.1 (Field/Mission Secured Cargo) of ITOP 1-2-601. The pallets shall show continued ease of opening per 4.3.2, no damage to the projectiles, no stress cracks, and no deformations.

4.6.4 Drop evaluation. Perform the following tests using six (6) conditioned (for 24 hours minimum) test pallets (2 hot, 2 cold, 2 ambient). After each test is performed the pallets shall show continued ease of opening per 4.3.2, no damage to the projectiles, no stress cracks, and only minor deformations.

4.6.4.1 Edgewise drop. The pallets shall be subjected to the following test. Each pallet shall be supported on one end of its base on a block approximately five inches high and placed at right angles to the skids. The opposite end of the pallet shall be raised and allowed to drop freely from a height of 12 inches onto a steel surface. This is then repeated at the opposite end of the pallet. Inspect pallet for cracks and deformations.

4.6.4.2 Cornerwise drop. Each loaded pallet shall be supported at the corner of its base on a block approximately five inches high. A block approximately 12 inches high shall be placed under the other corner of the same end of the pallet. The opposite end of the pallet shall be raised to a height of 12 inches and allowed to fall freely onto a steel surface. This test shall be repeated on the diagonally opposite corner. Inspect pallet for cracks and deformations.

4.6.4.3 Rollover drop. The loaded pallet shall be tipped slowly sideways until it falls freely and by its own weight from the base to the side, side to top, top to the other side, and the other side to the base, thus completing one revolution on a steel surface. Inspect pallet for cracks and deformations.

4.6.5 Pendulum impact. The loaded pallet shall be freely suspended by chains or cables and swung as a pendulum against any unyielding, steel, flat and vertical barrier. The pallet shall be pulled back from the barrier until the center of gravity is raised 20.5 inches, then released and allowed to swing freely against the barrier. This test shall be applied to the short and long sides of the pallet. Inspect pallet for cracks and deformations.

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4.6.6 Loose cargo vibration. The six (6) conditioned (for 24 hours minimum) test pallets (2 hot, 2 cold, 2 ambient) shall be subjected to the secured cargo vibration test. The loaded pallet shall be placed on the steel mounting surface (table) of the vibration equipment, in the normal shipping configuration (on skids). The pallet shall not be restrained during vibration except by a fence attached to the test surface to prevent the pallet from falling off the table. The table frequency shall be adjusted to assure that one edge of the pallet leaves the table not less than 3/16 inch on each cycle. Run the test at approximately 200 rpm for two hours, in accordance with FED-STD-101, Method 5019. The pallets shall show continued ease of opening per 4.3.2, no damage to the projectiles, no stress cracks, and no deformations.

4.6.7 Forty foot drop. The three pallets loaded with live projectiles shall be dropped from an height of forty (40) feet, onto a rigid horizontal impact surface. At each temperature only one pallet shall be dropped at an orientation of base down for cold, horizontal on the end for hot and horizontal on the side for ambient. Acceptance criteria shall be that no energetic reaction of any explosive or propellant occurs and that the projectiles are safe to collect and dispose.

4.6.8 Railcar impact test. Simulate boxcar coupling effect when projectiles are transported by railroad, in accordance with EVT-TP-1-86. This test is usually conducted at the US Army Defense Ammunition Center (USADAC). In this test, eighty (80) pallets are loaded with inert projectiles filling one row down the length of a box rail car. Twelve (12) of these pallets must be the test pallets. The rest of the sixty eight (68) pallets may be wood pallets. The loaded box car is then brought to a velocity of 4 mph and impacted into a parked train of box cars. This test is repeated at 6 mph, 8 mph and reverse 8 mph. Acceptance criteria after each test is performed is that the pallets show continued ease of opening per 4.3.2, no damage to the projectiles, no broken parts, no bending of components, no stress cracks, and no deformations.

4.6.9 Temperature shock test. Two (2) loaded pallets shall be subjected to the temperature shock test, in accordance with Method 503.3, Procedure I of MIL-STD-810. After each test is performed the pallets shall show continued ease of opening per 4.3.2, no deformations, and no stress cracks.

4.6.10 Salt spray test. Two (2) pallets should pass a 5 percent salt spray test for a minimum of 336 hours in accordance with ASTM B117. The pallet shall show no more than a trace of rusting per ASTM D610 and all moving parts shall function fully after test.

4.6.11 Chemicals, petroleum, oil, and lubricants verification. Satisfy the chemicals, petroleum, oil, and lubricants evaluation by one of the following methods.

4.6.11.1 Chemicals, petroleum, oil, and lubricants evaluation. Measure the physical properties of all the test sample materials. Submerge samples of the non-metallic materials used in the construction of the pallet to each of the chemicals, petroleum, oil and lubricants listed in Table VI for not less than 20 hours at $73^0 \pm 4^0$F and 50±5 percent humidity. Remove the samples from the liquids, allow excess fluid to drain, and the samples to dry for at least 12 hours at ambient temperature. Inspect for evidence of chemical reaction. Subject these material samples to the temperature test at both the high and low temperatures. Inspect for evidence of material degradation or damage. Compare the physical properties of tested materials to their pre-test property values. Any degradation is evaluation failure.

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4.6.11.2 Chemicals, petroleum, oil, and lubricants evaluation alternative. Verify pallet non-metallic component material compatibility with the list of chemicals, petroleum, oil, and lubricants listed in Table VI by laboratory chemical analysis or evaluation.

Table VI. Chemicals, petroleum, oil, and lubricants

Dry cleaning solvent, ASTM D235	Hydraulic fluid, fire-resistant, SAE J1447
Lubricant, LSA 1/	Hydraulic fluid, petroleum, ANSI B93.63M
Lubricating oil, ANSI/SEA AMS 3054A	Antifreeze, ethylene glycol, SAE J1034
Cleaner, lubricant, preservative, MIL-PRF-63460	Lubricating oil, ASTM D 4175
Gasoline, automotive combat, SAE J312	Turbine fuel, ASTM D1655
Diesel fuel, SAE J313	Lubricating oil, SAE J300
Insect repellent, ANSI K62.344-1993	
Chemical agent decontaminate, DS2 2/	
Chemical agent decontaminate, STB 3/	

1/ Lubricant, LSA, consists of the following materials in the proportions by weight indicated: Lithium stearate, 8.0±0.3 percent; lubricant-grade bis(2-ethylhexyl)sebacate, 89±1.0 percent; diisopropyl phosphite, 1.0±0.2 percent; 2,6 di-tertary butyl-p-crsol, 0.5±0.1 percent; oil-free barium dinonylnaphthalene sulfonate, 1.5±0.3 percent.

2/ Chemical agent decontaminate, DS2, consists of the following materials in the proportions by weight indicated: Diethylenetriamine, 69-71 percent; sodium hydroxide (ACS grade except the sodium carbonate content is no greater than 0.5 percent by weight), 1.9-2.1 percent; and ethylene glycol monomethyl ether, the remaining weight.

3/ Chemical agent decontaminate, STB, consists of the following materials in the proportions by weight indicated: calcium oxide, 6.6±0.1 percent: and bleaching powder (chlorinated lime having a maximum moisture content of 1.0 percent by weight (Do not substitute calcium hypochlorite for the bleaching powder), the remaining weight.

4.6.12. Sand demonstration. Perform this test after the salt spray and fungus test. One (1) pallet shall be subjected to the Blowing Sand test in accordance with Method 510.3, Procedure II of MIL-STD-810. Acceptance shall be in accordance with Method 510.3, paragraph I-4.1 of MIL-STD-810.

4.6.13 Dust demonstration. Perform this test after the salt spray and fungus tests. One (1) pallet shall be subjected to the Blowing Dust test in accordance with Method 510.3, Procedure I of MIL-STD-810. Acceptance shall be in accordance with Method 510.3, paragraph I-4.1 of MIL-STD-810.

4.6.14 Solar (UV) radiation evaluation. One (1) empty pallet shall be subjected to Solar Radiation test in accordance with Method 505.3, Procedure II, of MIL-STD-810. Acceptance shall be in accordance with Method 505.3, Procedure II, paragraph I-4.1 of MIL-STD-810.

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4.6.15 Fungus evaluation. One (1) empty pallet shall be subjected to the fungus test Method 508.4 of MIL-STD-810. Acceptance shall be in accordance with Method 508.4, paragraph I-4.1 of MIL-STD-810.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements will be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. These pallets covered by this specification are military unique because they are used for 155mm projectiles only and have no commercial application. The pallet is used to store and transport up to eight (8) 155mm artillery projectiles of varying lengths.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification and any amendment.
b. Issue of the DODISS to be cited in the solicitation and, if required, the specific issue of the original documents referenced (see 2.2).
c. When a first article is required (see 3.1, 4.1, 6.3 and 6.5).
d. What conformance verification is necessary (see 4.1.2.2 and 6.4).
e. Packaging requirements(see 5.1).
f. Provisions for Government Furnished Material (GFM) projectiles (see 4.2.3.1 and 6.9)
g. Length and type of projectile being procured.

6.3 First article. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.4 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: Contractor past performance, government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting

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documents should indicate those tests desired from Table I and their designated frequency based on a risk assessment for the procurement.

6.5 Drawings. For reference only, items produced using the following approved Technical Data Packages (TDPs) meet all the requirements and are approved for use with 155mmm projectiles:

12914619	Packing & Marking 155mm Projectile Pallet
8837839	Pallet for Projectile,155mm, HE & GB, Hollow Base Type
9362569	Packing and Marking for Pallet, Wood, Ammunition, For Projectile, 155mm

The current pallet for new artillery projectiles is the Field Artillery Projectile Pallet (FAPP), TDP number 12914619. The FAPP meets all Type I US Army requirements in this specification. The older wood pallet design, TDP number 8837839 and 9362569 meets Type II requirements which means it does not meet NBC decontaminability, non-flammability, repalletization, but may be used at the discretion of the procurement contracting officer. The contractor may use the information contained in the TDP as guidance for its own design. If the contractor chooses to build to the current approved TDPs, the Contracting officer may waive the testing required to meet First Article performance test requirements of this specification.

Copies of this drawing package are available from US Army Armament Research, Development, and Engineering Center ARDEC, AMSTA-AR-QAD, Picatinny Arsenal, NJ 07806-5000 or http://www.pica.army.mil.

6.6 Government loaned equipment. Contact the contracting officer for the on-going procurement to arrange for use of government loaned equipment.

6.7 Government technical point of contact: US Army Armament Research, Development, and Engineering Center (ARDEC), AMSTA-AR-WEP, Picatinny Arsenal, NJ 07806-5000

6.8 Recommended test facilities.

a. Yuma Proving Ground (YPG), Yuma, Arizona – Sequential Rough Handling testing.

b. U.S. Army Test & Evaluation Command (TECOM), Aberdeen Proving Ground, MD – Sequential Rough Handling testing.

c. U.S. Army White Sands Missile Range, White Sands Missile Range, New Mexico – Sequential Rough Handling testing

d. Combat System Test Activity (CSTA), Aberdeen Proving Grounds, MD – Live testing, system evaluation

e. US ARMY Defense Ammunition Center (USADAC), 1 C Tree Road, Bldg. 35, McAlester,

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Oklahoma 74501-9053 - unitization procedures and pallet testing

f. U.S. Army Dugway Proving Ground, Dugway, Utah – Nuclear, Biological, and Chemical Survivability

6.9 Government furnished material (GFM) projectiles. At the discretion of the procurement contracting officer, the government will provide to the contractor inert or live rounds for testing. The following table shows the quantities of inert or live rounds that are needed to conduct testing. For TYPE II and III pallets, the M864 projectile will be the test projectile. For TYPE I Style A pallets, the M107 projectile will be the test projectile. For TYPE I Style B, C and D pallets, the M864 projectile will be the test projectile. At the discretion of the procurement contracting officer, the specific round being procured may be used as the test projectile. The specific round firing procedure will be listed in the procurement contract.

Projectile	Quantity	Projectile Interface Drawings
M107	24 rounds	Figures 1, 2, 3.
M864	24 rounds	Figures 10, 11, 12.

6.10 Subject term (key word) listing.

Ammunition
Testing
Packaging

6.11 Definitions.

Damage: Anything that would impair proper function of the pallet, does not include minor scratches or dents.

Excessive force: Any force that is greater than the average soldier is capable of manually.

Free as possible: The least amount possible at no cost or reduction in performance.

Special tools: Any tool that is specially designed for the pallet, other than what is commonly found on the vehicle or in the field.

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APPENDIX A

A.1 SCOPE

A.1.1 SCOPE. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS. Not Applicable

A.3 PROCEDURE.

PROCEDURE FOR FIRING THE M864 PROJECTILE

1. Ammunition:
   Projectile, 155mm, HE, M864 w/live Base Burner Assembly
   Charge, Propelling, 155mm, M203A1
   Charge, Self-Registration
   Fuze, M577 MTSQ
   Primer, Percussion, M82
   Primer, Electric, M52E1B2

2. Weapon
   Carriage M198
   Recoil M45
   Gun M199
   Tube M199 60 percent estimated remaining life

3. Equipment, personnel and facilities
   Chamber, conditioning
   Camera, syncroballistic (Smear)
   Gage, Copper Crusher, M11
   Instrument, observer
   HAWK Doppler radar
   Transit, survey
   Stopwatch
   Gun Position, Kofa Firing Range, U.S. Army Yuma Proving Ground
   Gun Crew
   3-observer Geodetic Crew (to observe and determine flight times and impact range/deflections)
   EOD Crew (to flag and identify each ground impact test projectile for accurate geodetic measurements)
   Meterological data during firing.

4. Prior to firing, all projectiles are to be weighed with M577A1 MTSQ Fuzes. Condition eight projectiles at –45F (with fuzes at –25F) and eight projectiles at 145F. Specific round by round data for all projectiles are to be observed. M864 projectiles are to be fired with a M203A1, Zone 8

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Propelling Charge for explusion onto a prepared impact field. Three down range observers shall observe terminal ballistic data and maximum chamber pressure shall be observed using two Copper Crusher gages per round. After firing for explusion, the test projectiles are to be evaluated for proper grenade functioning by walking the impact field to identify dud grenades. In addition, holes are to be counted to determine the number of proper functioning grenades. After evaluating the dud grenades, they are to be destroyed in place.

5. Safety precautions in handling and testing: Standard procedures with the handling and testing of live ammunition must be observed. All firings and testing shall be performed remotely from behind a barricade with all personnel under suitable cover. Suitable hearing protection is to be worn during firings. Inspect projectiles carefully before chambering. Do not fire any damaged projectiles.

6. Information to be observed:

   Projectile Lot Number.
   Propelling Charge Lot Number.
   Fuze Lot Number.
   Meterological data during firing.
   Projectile weights.
   Smear photos at 20m forward of muzzle (metal parts separation).
   Any sign of premature functioning.
   Time of Flight.
   Height of Burst.
   Recovery/Examination of Ogives to verify proper ejection and submunition function (after every 8 rounds).

7. Pass/fail criteria shall be proper functioning of the M864 projectile.

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APPENDIX B

B.1 SCOPE

B.1.1 SCOPE. This appendix is a mandatory part of the specification. The information contained herein is intended for compliance.

B.2 APPLICABLE DOCUMENTS. Not Applicable

B.3 PROCEDURE.

PROCEDURE FOR FIRING THE M107 PROJECTILE

1. Ammunition:
   Projectile, 155mm, HE, M107
   Charge, Propelling, 155mm, M119A1
   Charge, Self-Registration
   Fuze, M557 PD
   Primer, Percussion, M82

2. Weapon
   Carriage M198
   Recoil M45
   Gun M199
   Tube M199 60 percent estimated remaining life

3. Equipment, personnel and facilities
   Chamber, conditioning
   Camera, syncroballistic (Smear)
   Gage, Copper Crusher, M11
   Instrument, observer
   HAWK Doppler radar
   Transit, survey
   Stopwatch
   Gun Position, Kofa Firing Range, U.S. Army Yuma Proving Ground
   Gun Crew
   3-observer Geodetic Crew (to observe and determine flight times and impact range/deflections)
   EOD Crew Meterological data during firing.

4. Prior to firing, all projectiles are to be weighed with M564 MTSQ Fuzes. Condition eight projectiles at –45F (with fuzes at –25F) and eight projectiles at 145F. Specific round by round data for all projectiles are to be observed. M107 projectiles are to be fired with a M119A1, Zone 8 Propelling Charge for impact onto a prepared impact field. Three down range observers shall observe terminal ballistic data and maximum chamber pressure shall be observed using two Copper Crusher gages per round.

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5. Safety precautions in handling and testing: Standard procedures with the handling and testing of live ammunition must be observed. All firings and testing shall be performed remotely from behind a barricade with all personnel under suitable cover. Suitable hearing protection is to be worn during firings. Inspect projectiles carefully before chambering. Do not fire any damaged projectiles.

6. Information to be observed:

   Projectile Lot Number.
   Propelling Charge Lot Number.
   Fuze Lot Number.
   Meterological data during firing.
   Projectile weights.
   Smear photos at 20m forward of muzzle (metal parts separation).
   Any sign of premature functioning.
   Time of Flight.

7. Pass/fail criteria shall be proper functioning of the M107 projectile.

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6.12 Changes from previous issue. Marginal notation are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 8140-0214)

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced documents(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYYYMMDD)
	MIL-PRF-32049 (AR)	23 SEPTEMBER 1999

DOCUMENT TITLE
PALLET, FIELD ARTILLERY PROJECTILE

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code)*	d. TELEPHONE (*Include Area Code)* (1) Commercial (2) DSN (*if applicable)*	7. DATE SUBMITTED (*YYYYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (*Include Area Code)*	
U.S. Army TACOM-ARDEC Standardization Team	(1) Commercial (973) 724-5822	(2) DSN 880-5822
c. ADDRESS (*Include Zip Code)* Attn; AMSTA-AR-QAW-E Picatinny Arsenal, NJ 07806-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Standardization Program Office (DLSC-LM)** **8725 John J. Kingman Road, Suite 2533** **Fort Belvoir, Virginia 22060-6221** **Telephone (703) 767-6888 DSN 427-6888**	

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